PAAL PHARMA MACHINERY PVT. LTD.

PRODUCTS CATALOG

Pharmaceuticals, Chemicals, Food & Allied Processing Machineries

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WE ARE SERVING

- Tablet Section
- Oncology Section
- Semi-solid Section
- Liquid Section
- Injectable Section
- Capsule Section
- OSD Section
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- Chemical
- Orgaanic/ Pesticide
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- Pigment
- Food
- Intermediate

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WELCOME TO PAAL PHARMA MACHINERY PVT. LTD.

“Our Goal is to extent our market & impact by gaining the trust & certainty of clients all over the place. We are reslove to serve our esteemed customers by providing abnormal state innovation promotion the best quality items. We have confidence in conveying brief & the best client administration.”

INTEGRATED GRANULATION SUITE:-

- Rapid Mixer Granulator
- Fluid Bed Processor
- Vacuum Transfer System
- Co - Mill
- Blender

The Integrated Granulation Suite is designed as a closed and sequential processing system for wet granulation, drying, milling, and blending of pharmaceutical materials. The system integrates multiple equipment through vacuum transfer and pipelines to ensure minimal manual handling, reduced contamination, and consistent product quality.

- All contact parts made of S.S.316L/304 with isolated control panel
- Available in GMP compliant models
- PLC-based control panel for automated operation/ 21 CFR compliance
- Modular design for easy operation and maintainance
- CIP and SIP provision for cleaning
- **Capacity Range** - 15 L to 1600 L

Models	PPMPL-15	PPMPL-30	PPMPL-90	PPM-180
Bowl Volume in Ltrs	15	30	90	180
Batch Capacity (KG)	5	10	30	60
Motor HP	5	7.5	10	15
Heating Capacity in kW	9	12	18	36
Drying Temperature °C	30-80	30-80	30-80	30-80
Height in mm	1400	1800	2000	3500
Width in mm	1050	1050	1500	1500
Length in mm	2550	2850	3850	4950

FLUID BED PROCESSOR

A fluid bed processor is a versatile, high-efficiency machine used in pharmaceutical and chemical industries to transform powders into granules (granulation), dry materials, or apply coatings. By suspending particles in an upward-moving, heated gas stream (fluidization), it allows for rapid heat and mass transfer.
Combines drying, top-spray/bottom-spray granulation, and coating (e.g., pellet coating) in a single unit.

- **Versatile Functionality:** Performs multiple functions in a single pot, including drying, spray granulation, and coating (top and bottom spray/Wurster).
- **GMP Compliance & Construction:** All contact parts are made of high-quality stainless steel (SS 316L) with non-contact parts in SS 304, complying with stringent cGMP standards.
- **Process Efficiency:** High-speed fluidization allows for rapid drying and efficient heat exchange.
- **Automation & Control:** Features PLC (Programmable Logic Controller) systems with HMI (Human Machine Interface) touchscreens for precise control over parameters like spray rate, air flow, and temperature.
- **Process Visualization:** Transparent inspection windows allow operators to observe the fluidization process.
- **Safe Handling & Operation:** Equipped with pneumatic sealing for containers, explosion relief flaps, and automatic bag-shaking mechanisms for cleaning.
- **Advanced Spray System:** Utilizes top spray for granulation and bottom spray/Wurster systems for coating, using precise atomizing nozzles.
- **Environmental Control:** Inlet air can be filtered through HEPA filters to meet cleanroom standards
- **Capacity** - 5 KG to 500 KG

RAPID MIXER GRANULATOR

A Rapid Mixer Granulator (RMG), or high-shear mixer, is a crucial pharmaceutical machine that blends powders and creates granules in a single, fast, and efficient step. It uses a slow-speed impeller for mixing and a high-speed chopper to create uniform, free-flowing granules, significantly reducing processing time compared to traditional methods.

- **Rapid Processing & High Efficiency:** The combination of a main impeller and a high-speed chopper (1440/2880 RPM) ensures fast dry mixing and wet granulation, reducing processing time.
- **Construction & Hygiene:** All contact parts are typically constructed from stainless steel (SS 316L/304), designed for easy cleaning with mirror finishes. They are fully compliant with cGMP regulations.
- **Capacity Range** - 3 L to 1200 L

STARCH PASTE KETTLE

Starch Paste Kettle is used for Preparation of Starch Paste in Pharmaceutical Industries. Special Conical Shape ensures that the Heating of the Starch Paste is uniform and gets more surface area. Paste Kettle with Tilting Arrangement for Unloading the Paste to Container or Vessel, it ensures hygiene, easy cleaning, and efficient mixing. The kettle works by heating starch with additives, mixing thoroughly, and discharging the finished paste effortlessly.

- **Design & Construction:** Compact GMP compliant model with a hemispherical base, designed for optimal heat transfer and paste discharge.
- **Heating System:** Jacketed design for heating using steam, electricity, or oil, often insulated with glass wool to prevent heat loss..
- **Temperature Control**: Equipped with a digital temperature controller and indicator.
- **Capacity Range** - 20 L to 500 L

CO - MILL

A Co-Mill (conical mill) reduces particle size using low-shear, high-speed centrifugal force, driving material against a fixed conical screen with an impeller. Material is fed into the cone, where blades break it down and force particles through screen holes, ensuring uniform size, low heat generation, and low dust.

- **Versatility:** Suitable for wet and dry milling, de-lumping, and screening.
- **Uniform Particles:** Produces a tighter particle size distribution compared to other methods.
- **Low Impact Processing:** Gentle action ensures low heat generation, noise, and dust, essential for heat-sensitive products.
- **Efficient Design:** Easy to clean (often with quick-release, clamp-on designs)
- **Capacity -** 1 HP to 20 HP

MULTI - MILL

A multi mill is a versatile, high-speed industrial machine used for granulation, pulverizing, mixing, shredding, and chopping wet or dry materials. Primarily used in pharmaceuticals, it works by combining rotating swing hammers (with knife/impact edges) and screen, creating centrifugal force that results in controlled, efficient particle size reduction.

- **Variable Speed & Direction:** Equipped with a Variable Frequency Drive (VFD) to adjust speed and a switch to change blade rotation direction (forward for knives, reverse for impact).
- **High-Quality Construction:** Typically constructed from stainless steel (SS 316L or SS 304) for hygienic processing, with easy dismantling for cleaning.
- **Mobility & Compact Design**: Mounted on castor wheels for easy portability, with a compact footprint to save space.
- **Capacity Range** - 1 HP to 10 HP

VIBRO SIFTER

A vibro sifter, also known as a vibrating screen or safety screener, is an industrial machine used to separate, grade, and scale materials like powders, granules, and liquids based on their particle size.

- **Multi-Deck Flexibility**: Supports double or triple-deck screen arrangements, enabling multiple particle size separations in a single pass.
- **Easy Maintenance & Mobility:** Designed for easy disassembly and assembly for quick, thorough cleaning and screen changes, reducing downtime. Often mounted on castor wheels for portability.
- **Capacity Range -** 12" to 72"

CONTA BLENDER

A Conta Blender (or Bin Blender) is a high-efficiency pharmaceutical blending system designed for dust-free mixing of powders and granules, acting as a cGMP-compliant alternative to conventional blenders. It operates by tumbling materials within a sealed container (bin) at a diagonal, eccentric angle, ensuring homogeneous mixing, increased production efficiency, and reduced handling between processing stages.

In the context of industrial mixing, the Bin (specifically an Intermediate Bulk Container or IBC) is the detachable vessel that holds the materials throughout the entire manufacturing process—from charging and blending to storage and discharge

- Interchangeable bins allow multiple batch handling with one machine.
- Space-saving design with mobile bin system.
- Eliminates cross-contamination by dedicated bin usage.
- Reduced manpower requirement with automated lifting and docking.
- Uniform blending achieved through bin rotation and tumbling.
- Meets cGMP standards with dust-free closed operation.
- Modern units may include features like Variable Frequency Drives (VFD) for speed control, digital timers, and optional Programmable Logic Control (PLC) systems with Human Machine Interface (HMI) for automated operation and visual display.
- **Capacity -** 5 L to 10,000 L

AUTO-COATER

An auto coater is a high-efficiency machine used, mainly in the pharmaceutical and food industries, to apply uniform film or sugar coatings to tablets, pills, and granules. They optimize production with automated, PLC-controlled spray systems for consistent quality, ensuring taste masking, stability, and precise, rapid coating in compliance with cGMP standards.

- **Automation & Control:** Fully automatic PLC-based systems manage spray rate, atomization pressure, air volume, and temperature, reducing human error and ensuring consistent, high-quality tablet coating.
- **Perforated Pan Design:** The rotating drum is designed with specialized baffles (e.g., shark fins) to minimize tablet breakage and enhance coating uniformity.
- **Advanced Spray System:** High-precision spray guns with adjustable arms ensure accurate coating application, with peristaltic pumps for precise solution dosing.
- **cGMP Compliance:** Machines are constructed to comply with Good Manufacturing Practices, featuring stainless steel (SS316L/304) contact parts, mirror-finished surfaces.
- **Efficient Drying & Air Handling**: Features an inlet air handling unit (AHU) with HEPA filtration and dehumidification, along with an exhaust system to manage temperature, pressure, and dust.
- **Capacity** - 6, 12, 24, 36, 48, 60, 96, 192 Trays

COATING PAN

- **Construction & Design:** Manufactured in compliance with cGMP guidelines, featuring polished stainless steel (SS 316L or 304) contact parts for hygiene. Often designed with welded baffles for optimal tablet mixing.
- **Drying System:** Equipped with a hot air blower featuring a thermostat control, allowing temperatures to be regulated for efficient,, rapid drying.
- **Operational Control:** Includes variable speed drives (VFD) to adjust rotation speed and adjustable tilt angles (up to 45 degree) to optimize the movement of different products.
- **Spraying System:** Optional high-precision spraying systems (nozzles) with peristaltic pumps allow for consistent application of coating solutions.
- **Capacity Range -** 6", 12", 18", 24", 36", 48", 60", 72".

MANUFACTURING PLANT

The Ointment Manufacturing Plant is designed for producing ointments, creams, lotions, and emulsions.
A complete ointment manufacturing plant generally consists of three main vessel types:

- **Main Manufacturing Vessel:** The central unit where final homogenization occurs. It features a top-entry anchor agitator with Teflon scrapers to prevent wall-caking and a bottom-entry homogenizer for particle size reduction, often below 5-10 microns.
- **Wax Phase Vessel**: A jacketed vessel used to melt and mix oil- or wax-based ingredients. It typically uses a high-speed saw-tooth cutter stirrer to ensure uniform solubility at high temperatures.
- **Water Phase Vessel:** Used for aqueous ingredients, this jacketed vessel employs a propeller-type stirrer with baffles to prevent vortex formation while heating or cooling the solution.
- **Storage Vessel:** A portable, skid-mounted tank with a conical bottom, used to hold the finished product before it is transferred to a filling machine.

FILLING MACHINE

Liquid filling machines offer high precision, versatility, and efficiency, featuring stainless steel (SS316L/SS304) construction for hygiene, diving nozzles for foam-free filling, and anti-drip technology to prevent wastage. Key features include PLC controls with HMI touchscreens, adjustable speed, and "no bottle – no fill" sensors.

Types : -

- **Automatic Tube Filling Machines:** These handle the entire cycle—from loading and orientation to filling and ejection—without manual intervention. High-speed models can process up to 120 tubes per minute.
- **Semi-Automatic Machines:** Require manual tube loading but automate the filling and sealing process. These are typically more affordable and better suited for small-batch production.
- **Rotary vs. Linear:** Rotary machines use a rotating disc to move tubes between workstations, making them compact. Linear machines move tubes in a straight line and are often used for high-capacity lines.
- **Capacity** - 10 KG to 5,000 KG

MANUFACTURING PLANT

The Liquid Manufacturing Plant in the Pharmaceutical Industry is mainly used to produce Oral Liquid or Sugar Syrup. A complete liquid/ syrup manufacturing plant generally consists of three main vessel types:

- **Sugar Melting Vessel:** This initial vessel is used to dissolve sugar and other primary ingredients into water. It is usually equipped with a high-speed stirrer or propeller-type agitator and a heating jacket (steam, electric, or hot water) to facilitate melting.
- **Main Manufacturing Vessel:** After initial melting and filtration, the solution is transferred here for final formulation. It often features a bottom-mounted propeller to prevent lump formation and an inline homogenizer to reduce particle size (typically below 5 microns) for a smooth texture.
- **Storage Vessel:** Once the syrup reaches the desired viscosity and composition, it is moved to a storage tank. This vessel keeps the product stable and may include a low-speed stirrer to maintain uniformity until it is ready for the filling line.

FILLING MACHINE

Capacity: Determine the volume of liquid you need to fill per unit of time. Liquid filling machines come in various capacities, ranging from small-scale tabletop models to high-speed industrial machines.
Range - 10 L to 10,000 L
Versatility: Consider the types of liquids you'll be filling, whether it's thin liquids like water or viscous substances like oils and syrups. Choose a machine capable of handling a wide range of liquid consistency without compromising efficiency.
Ease of Cleaning and Maintenance: Opt for machines with easy-to-clean components and minimal maintenance requirements. This ensures smooth operation and prolongs the lifespan of the equipment.
Automation Level: Depending on your production requirements and budget, decide between manual, semi-automatic, or fully automatic liquid filling machines. Automation can significantly enhance productivity and reduce labor costs.

STERILE MANUFACTURING VESSEL

Sterile manufacturing vessels are specialized, fully enclosed stainless steel tanks (15 L to 10,000 L+) designed for aseptic processing in the pharmaceutical industry. They ensure product sterility and safety for parenteral, ophthalmic, and liquid products, incorporating.

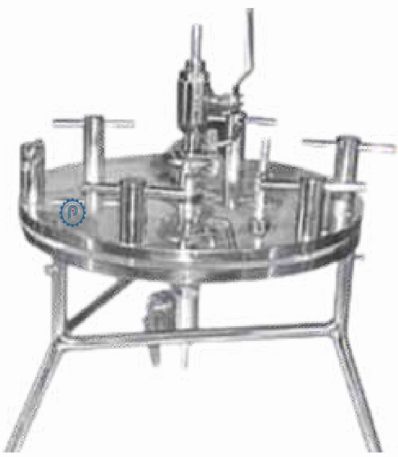
- **Sterilization Systems:** Equipped with CIP (spray balls) and SIP (steam-in-place) ports to thoroughly sterilize the vessel and transfer lines.
- **Zero Dead Leg Design:** Employs aseptic diaphragm valves to prevent "dead legs" where fluid can stagnate and microbes can breed.
- **Mixing Mechanisms:** Utilizes bottom-mounted magnetic mixers to eliminate potential leakage from mechanical seals, often offering variable speed control.
- **Capacity -** 15 L to 2,000 L.

PRESSURE VESSEL

Pressure vessels are specialized containers designed to hold gases or liquids at pressures significantly higher or lower than the ambient atmosphere.

- **Structural Design:** Primarily cylindrical, spherical, or conical to optimize pressure distribution.
- **Vessel Heads:** End closures that include ellipsoidal, hemispherical, or torispherical shapes to safely manage internal stresses.
- **Safety Devices:** Equipped with safety valves or rupture discs to prevent hazardous over pressure.
- **Capacity** - 25 L to 500 L.

MEMBRANE HOLDER FILTER



In the context of industrial filtration, a membrane holder is a specialized component within a filter press that utilizes a flexible, impermeable surface to mechanically squeeze filter cakes.

- **Holder Base:** This is the main body of the filter holder, usually made of plastic or stainless steel. It provides structural support and houses the other components.
- **Filter Support**: The filter support is a porous disc or screen that holds the membrane filter in place. It ensures that the filter remains flat and doesn't collapse during filtration.
- **Sealing O-ring:** The O-ring provides a tight seal between the filter holder base and the filter support, preventing leakage and maintaining the integrity of the filtration process.

FLUID BED DRYER

A fluid bed dryer (FBD) is an industrial machine that uses high-velocity hot air to dry granular, powder, or crystalline materials by suspending them in a fluidized state. It offers superior, uniform heat transfer and fast drying times (20-40 minutes) compared to conventional methods, making it ideal for pharmaceutical, chemical, and food industries.

- The Same Temperature Continuously remains.
- Material is dried from particle to particle
- Simple operation with, automated, PLC controls and quick-release mechanisms for easy cleaning.
- Designed with stainless steel (SS316L/ 304) to meet cGMP standards, featuring closed-loop handling to minimize contamination.
- **Safety Features:** Includes explosion vents (rupture discs) at the rear, explosion-proof construction for organic solvents, and earthing points to eliminate static charges.
- **PLC Automation:** Features fully automatic operations, including Programmable Logic Controls (PLC) with HMI touchscreens for process monitoring and data recording (e.g., airflow, temperature, batch reports).
- **Capacity Range** :- 5 KG to 500 KG

PRODUCT CONTAINER

Fluid Bed Dryer (FBD) product containers are critical pharmaceutical components manufactured to ensure fast, uniform drying, granulation, or coating, typically designed for cGMP (current Good Manufacturing Practices) compliance. They are engineered for high-pressure resistance and efficient air-particle contact.

- **Material of Construction (MOC):** Generally AISI SS316 or SS316L for all contact parts, with non-contact parts made of SS304.
- **Design:** Conical shaped with a tapered bottom to ensure no dead spaces and optimal fluidization.
- **Sealing:** Food-grade silicone inflatable seals are used to ensure leak-proof, airtight operation under pressure.

AIR TRAY DRYER

An air tray dryer (or cabinet dryer) is an enclosed, insulated chamber used for batch drying, where materials placed on stacked trays are dried through forced hot air circulation. Widely used in pharmaceuticals, food, and chemicals, it ensures uniform, efficient drying. Key features include temperature control, stainless steel construction, and GMP compliance for hygienic processing.

- **Mechanism**: A blower fan circulates heated air through heaters and baffles, creating a controlled, high-velocity airflow over the trays.
- **Loading:** Materials are loaded onto trays, which are typically stacked on trolleys, allowing easy movement in and out of the chamber.
- **Process Control:** Equipped with a control panel to monitor temperature and drying time, ensuring consistent results.
- **Capacity** - 6, 12, 24, 36, 48, 60, 96, 192 Trays

VACUUM TRY DRYER

A vacuum tray dryer is a batch-operated industrial machine used to dry heat-sensitive, delicate, or hazardous materials at lower temperatures by creating a vacuum, which lowers the boiling point of liquids. It is ideal for pharmaceuticals, chemicals, and food, reducing oxidation and preventing damage to the product while efficiently removing moisture using conductive, heated trays

- Stainless steel construction (SS 316L/304)
- Easily removable hollow pad type heating shelves
- Dedicated fluid heating and circulating systems
- Thermal efficiency of 60 to 80 per cent
- Vacuum break valve provided on Chamber
- High quality interlocking for doors to chamber
- Uniform heating flow
- **Capacity** - 6, 12, 24, 36, 48, 60, 96, 192 Trays

ROTOCONE VACUUM DRYER

The Rotocone Vacuum Dryer (RCVD) is an advanced pharmaceutical and chemical processing unit designed for efficient, gentle drying of heat-sensitive, toxic, or oxidized-prone materials under vacuum.

- **Vacuum Drying & Solvent Recovery:** Operates under high vacuum to lower boiling points, enabling low-temperature drying and full solvent recovery.
- **GMP Compliance:** Designed for high hygienic standards (c GMP) with product-contact parts made of SS 316L, ensuring suitability for pharmaceutical/food production.
- **Integrated Lump Breakers:** Equipped with internal lump breakers that continuously break up agglomerates during rotating action to produce fine, dry powder.
- **Efficient Drying Process:** Utilizes rotary tumbling and uniform heating (via steam/hot water jacket) for efficient heat transfer, reducing overall drying time
- **Capacity** - 50 l to 5000 L

AGITATED NUTSHE FILTER & DRYER

An Agitated Nutsche Filter Dryer (ANFD) is a closed-system vessel used in pharmaceutical and chemical industries for high-purity solid-liquid separation, combining filtration, washing, and drying in one unit. It operates under pressure or vacuum, utilizing an agitator to speed up processing and discharge dry powder.

- **Filtration:** Slurry is loaded, and a pressurized gas (e.g., Nitrogen) forces the liquid through a filter cloth/plate, forming a solid cake on top.
- **Washing:** Wash liquid is added and mixed with the solids using the agitator to remove impurities, followed by further filtration.
- **Drying:** The vessel is heated (often using a jacket or heated agitator) while maintaining a vacuum. The agitator (rotating arms) rotates and moves vertically to mix, crack, and dry the cake efficiently.
- **Discharge:** Once dried, the agitator moves in a reverse direction to automatically discharge the product through a side valve.

ZERO HOLD-UP FILTER PRESS

Zero Hold Up Filter Press is extensively used in the chemical industry for the filtration of chemicals such as acids, alkalis, solvents, and other corrosive liquids.

- Horizontal Plate / Sparkler Filter Press is enclosed construction, preventing evaporation, oxidation, leakage and fumes escaping from product.
- Sparkler Filter plates are available in two sizes. Deep plate for large percentage of cake holding capacity and Shallow plate for small percentage of cake holding capacity.
- In the Sparkler Filter Press, The horizontal filter plate and even thickness of cake, prevents the cake dropping as well as cracking thus assuring better filtrate quality.
- Process: The unfiltered liquid is fed under positive pressure into the top of the shell. It moves down and spreads over the filter media on the plates.
- Filtration: The solid impurities are held on top of the filter media (paper/cloth), while the clear liquid passes through the central channel to the outlet.
- Maintenance: The swing bolt design allows for easy opening and cleaning of the plates.

NUTSCHE FILTER

Nutsche filtration is defined as a batch filtration technique that uses vacuum or pressure in a closed vessel. The pressure or vacuum is used in combination with a nutsche filter plate fitted with the appropriate filter media to force liquid pass through the media to form a solid bed, often referred to as pressure Nutsche filters. This type of filter offers many benefits like product isolation & operator safety that make nutsche filters or dryers mainly used in pharmaceuticals, chemicals & wastewater treatment.

- Nutsche filters are primarily designed for solid-liquid separation through filtration.
- They excel at removing solids from liquid suspensions, enabling the recovery of valuable products or the purification of solutions.
- Nutsche filters are dedicated to the filtration step, requiring additional downstream processing for drying or further treatment of the filtered solids.
- **Capacity** - 20 L to 5,000 L

OCTAGONAL BLENDER

Octagonal Blender equipment has an octagon shaped body with a rectangular central portion. It is equipped with baffles that enable fast and efficient mixing as well as rapid charging and discharge ports with suitable butterfly valves. The equipment is highly suitable for areas where gentle blending of dry granules or powder is required.

The blender operates on the principle of **gravity-based tumbling** . As the octagonal drum rotates slowly, the material inside repeatedly falls and cascades from all sides. Baffles (internal plates) are often added to break the flow pattern and increase randomness, ensuring thorough mixing with minimal friction and heat generation

- **Rotation Speed**: Operates at slow speeds, typically between **5 to 25 RPM**, to prevent particle degradation.
- **Working Volume**: Efficiency is highest when filled to roughly **two-thirds (60-75%)** of its total capacity.
- **Materials**: Usually constructed from **Stainless Steel 316L** for contact parts and **SS 304** for non-contact components to meet pharmaceutical standards (cGMP)
- **Capacity Range** - 5 L to 20 L

V - CONE BLENDER

A V-cone blender (or twin-shell blender) is a high-efficiency tumbling mixer designed for blending dry powders and granules, commonly used in pharmaceutical, food, and chemical industries. Its "V" shape allows materials to divide and converge repeatedly, creating homogeneous mixtures with minimal attrition on fragile granules.

- **Optimal Blending Action:** The V-shape, typically at a 75° to 90° angle, constantly splits and recombines materials for superior homogenization.
- Gentle Handling: Ideal for delicate, friable materials because it uses a low-impact tumbling mechanism rather than high-shear mixing.
- Hygienic Design & Easy Cleaning: Internal surfaces are smooth and often mirror-finished to prevent material buildup and ensure easy cleaning.
- **Complete Discharge:** The conical design allows for full discharge at the tip of the V with minimal product retention.
- **Capacity** - 5 L to 10,000 L

DOUBLE CONE BLENDER

A double cone blender is a high-efficiency tumble blender used for uniformly mixing dry powders, granules, and fragile materials, commonly in pharmaceutical, food, and chemical industries. It uses a conical, rotating body to achieve gentle, low-shear blending with easy discharge, typically filling 2/3 of its volume for optimal results.

- Double Cone shaped product container is for Ideal dry mixer & lubrication of granules for homogenization mixing of multiple batches in to single batch.
- Enclosed rigid drive with reduction gear box and motor.
- Product container having discharge with butterfly valve and a man hole.
- Fixed & removable baffles provided for lumps braking.
- **Capacity -** 5 L to 10,000 L

RIBBON BLENDER

A ribbon blender is an industrial, horizontal mixer comprising a U-shaped trough and a double helical agitator that creates a convective mixing pattern to thoroughly blend powders, granules, and pellets.

- The lid has 3 piece segment with fixed centre piece and hinged cover at either sides for charging of products
- The main shaft made of stainless steel on which the ribbons are welded
- The ribbons are made of stainless steel flats with suitable size and thickness depending up on the capacity.
- **Capacity -** 5 L to 10,000 L

MASS MIXER

Mass mixer granulators are designed for high-efficiency mixing and granulation of dry/wet powders, primarily in the pharmaceutical, food, and chemical industries. Key features include cGMP-compliant stainless steel (SS 316L/304) construction, horizontal paddle agitators, and specialized chopper blades for producing uniform granules with short batch times.

- Suitable for mixing of dry & wet material.
- The Mass Mixer Machine is basically mixing a assembly wherein the mixing stirrer is in horizontal position in the container & have a single speed, simple rotation.
- The stirrer rotates around it self at a very slow speed inside the bowl and thereby achieving intimate mixing of dry or wet materials of Tablet Granulation, Powder, Chemicals, Food & Confectionaries Materials.
- **Capacity** - 5 L to 10,000 L

DUST COLLECTOR

Dust collectors in the pharmaceutical industry are critical engineering controls designed to capture fine, toxic, or hazardous airborne powder during processes like tablet pressing, coating, granulation, and blending.

- Effectively removes fine dust particles, improving air quality and compliance with safety standards.
- Equipped with a filter-cleaning mechanism to maintain consistent performance without frequent manual intervention.
- **Capacity -** 1 HP to 10 HP

TOP ENTRY HOMOGENIZER

A homogenizer is a high-shear mixing device used to achieve uniform particle size reduction and consistent blending of liquids or semi-solids. It is widely used in pharmaceutical and process industries to ensure product stability and quality.

- Ensures rapid size reduction and uniform mixing for consistent output quality.
- Interchangeable stators (coarse/medium/fine) allow flexibility to meet specific process requirements.
- **Capacity** - 1 HP to 20 HP

TOP ENTRY STIRRER

Top entry stirrers, or agitators, are versatile, top-mounted industrial mixing devices utilized for blending, suspending, and homogenizing liquids in tanks, featuring motors mounted above the vessel driving a shaft with impellers.

- **Drive Types:** Options include direct drive for high speed or gear-reduced for high torque, handling viscous fluids.
- **Construction Materials:** Available in Stainless Steel (SS316L/SS304), Hastelloy, or with anti-corrosive coatings for chemical applications.
- **Capacity -** 1 HP to 20 HP

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